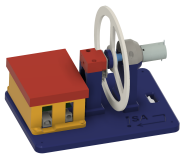
Quick start guide

DC motor control lab

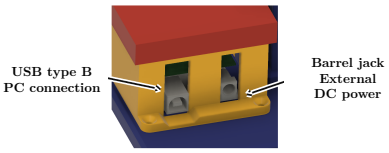
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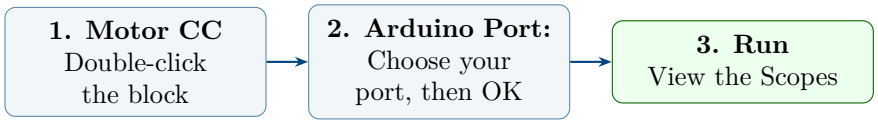
Starting point: the rig is assembled and the firmware is already installed on Arduino.

Required software: MATLAB with Simulink. Instrument Control Toolbox is the only additional toolbox required to use the rig. Arduino IDE, the Arduino support package and other toolboxes are not needed to communicate with the rig.

Check the installation: in the MATLAB *Command Window*, type `ver` and press Enter. **Instrument Control Toolbox** must appear in the list of installed products. If it is missing, follow the installation appendix at the end of this guide.



USB: connection to the computer. Barrel jack: external power.
Connect USB first, then the external power supply.



1. **Connect the rig.** Connect Arduino to the computer using USB. If using an external supply, connect it after USB: 8.4 V recommended, 9 V maximum. USB power alone gives a smaller speed range. Keep the motor shaft clear.

2. **Open the model.** In MATLAB, browse to `software/matlab/` and open the `MotorCC_V3_Rxxxx.slx` file for your release. Example: `MotorCC_V3_R2024b.slx` for R2024b.

3. **Edit the mask parameters.** With the simulation stopped, **double-click the central Motor CC block** (`Motor_CC`). Its parameter dialog, or *mask*, opens.

4. **Select the port and confirm.** In the **Arduino Port:** dropdown, choose the Arduino port and click **OK**. Examples: `COM3` (Windows), `/dev/cu.usbmodem...` (macOS), or `/dev/ttyACM0` (Linux). There is no need to open the mask editor or change internal blocks.

5. **Run the experiment.** Click **Run** in Simulink. Open the **Scope** blocks to view the signals. Position is in degrees and velocity in degrees per second. The supplied model applies an open-loop test signal; no feedback controller is connected.

6. **Finish.** Click **Stop** and wait for the motor to stop: after 1 s without commands, Arduino reduces PWM by 2 units every 10 ms until zero. Disconnect external power first, if connected, then USB.

Which port belongs to Arduino? Close the mask, unplug USB and reopen the mask to inspect the list. Close it, reconnect USB and open it again: the new port belongs to Arduino. Select it and click **OK**.

Connection problem? Close Arduino Serial Monitor and any other program using the port. Check the USB cable and select the port again in the mask. If serial blocks are missing, check that Instrument Control Toolbox is installed.

No firmware on Arduino? Follow the separate *Firmware installation* guide (`firmware_installation_isa.pdf`).

Appendix · Install Instrument Control Toolbox

Installation is performed from MATLAB and is needed only once on each computer.

1. Connect the computer to the Internet and open MATLAB.
2. On the **Home** tab, open **Add-Ons** and choose **Get Add-Ons**. The wording may vary by release.
3. Search for **Instrument Control Toolbox** and open the product published by **MathWorks**. Select the main toolbox, rather than a support package for specific instruments.
4. Click **Install**. If prompted, sign in with your MathWorks account associated with the university license and follow the installer instructions.
5. When installation finishes, return to MATLAB, type `ver` in the Command Window and check that **Instrument Control Toolbox** appears. Then continue with the steps on the first page.

About the license. Instrument Control Toolbox is a licensed MathWorks product; it is not generally available as a free toolbox. If it is included in your university license, you can install and use it without purchasing an additional license. If the installer does not offer the product, check that you have selected the correct account and license.

If installation is not offered: run the MATLAB installer for your installed release, select the same license and installation folder, and add **Instrument Control Toolbox** to the installed products.

Official instructions: MathWorks · Add products to an existing MATLAB installation.